

 INDIAN INSTITUTE OF TECHNOLOGY KHARAGPUR										Stamp / Signature of the Invigilator	
EXAMINATION (End Semester)										SEMESTER (Spring)	
Roll Number										Section	Name
Subject Number	C	S	3	0	2	0	2	Subject Name			Database Management Systems
Department / Center of the Student										Additional sheets	

Important Instructions and Guidelines for Students

1. You must occupy your seat as per the Examination Schedule/Sitting Plan.
2. Do not keep mobile phones or any similar electronic gadgets with you even in the switched off mode.
3. Loose papers, class notes, books or any such materials must not be in your possession, even if they are irrelevant to the subject you are taking examination.
4. Data book, codes, graph papers, relevant standard tables/charts or any other materials are allowed only when instructed by the paper-setter.
5. Use of instrument box, pencil box and non-programmable calculator is allowed during the examination. However, exchange of these items or any other papers (including question papers) is not permitted.
6. Write on both sides of the answer script and do not tear off any page. **Use last page(s) of the answer script for rough work.** Report to the invigilator if the answer script has torn or distorted page(s).
7. It is your responsibility to ensure that you have signed the Attendance Sheet. Keep your Admit Card/Identity Card on the desk for checking by the invigilator.
8. You may leave the examination hall for wash room or for drinking water for a very short period. Record your absence from the Examination Hall in the register provided. Smoking and the consumption of any kind of beverages are strictly prohibited inside the Examination Hall.
9. Do not leave the Examination Hall without submitting your answer script to the invigilator. **In any case, you are not allowed to take away the answer script with you.** After the completion of the examination, do not leave the seat until the invigilators collect all the answer scripts.
10. During the examination, either inside or outside the Examination Hall, gathering information from any kind of sources or exchanging information with others or any such attempt will be treated as 'unfair means'. Do not adopt unfair means and do not indulge in unseemly behavior.

Violation of any of the above instructions may lead to severe punishment.

Signature of the Student

To be filled in by the examiner											
Question Number	1	2	3	4	5	6	7	8	9	10	Total
Marks Obtained											
Marks obtained (in words)				Signature of the Examiner				Signature of the Scrutineer			

Instructions: Answer all FOUR questions. Time = 3hrs. Total marks = $4 \times 25 = 100$. Write your answers only in the space provided. Show all the solution steps. Answers without explanation will be penalised. The question paper has total 28 pages.

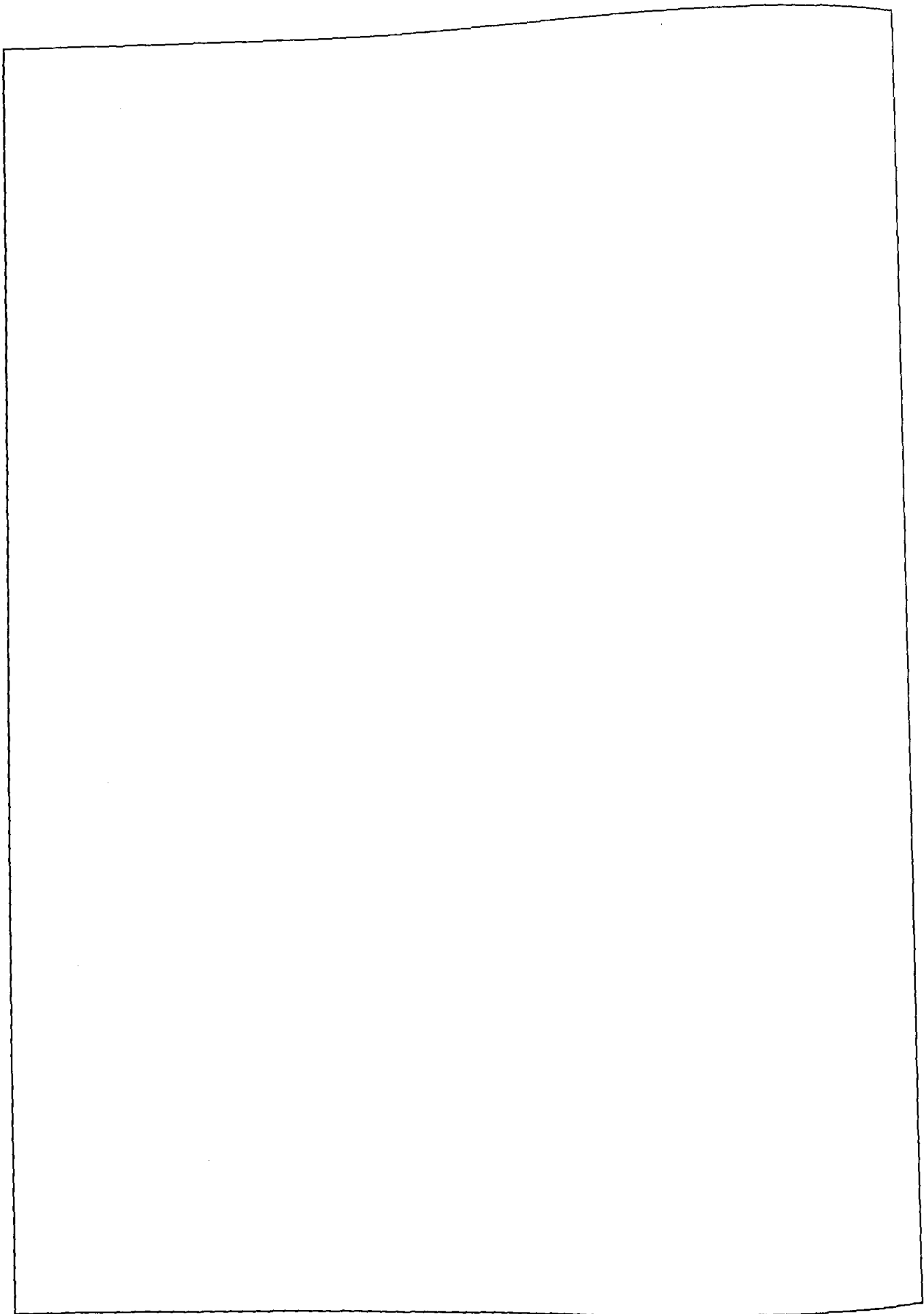
ROUGH WORK

1.(a) In view of B+ index tree, answer the following:

[3+5=8]

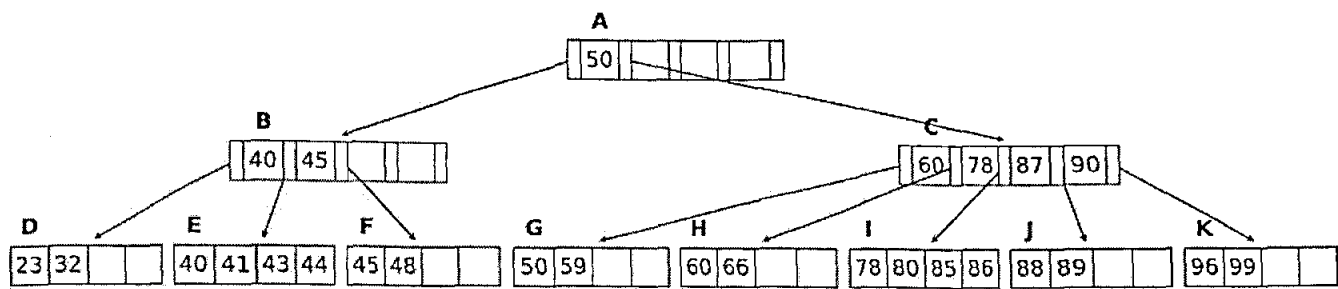
(i) With appropriate diagrams discuss the structure and salient features of internal and leaf nodes.

(ii) Compute the number of keys present in the internal node and leaf node for the following scenario: Size of the block = 1024 bytes, size of the key = 10 bytes, size of the node-pointer = 8 bytes and size of the record pointer = 12 bytes. Assume that each of the internal node or leaf node will fit into one block. How many block accesses are required to access a record from a file containing 10,00,000 records using the above mentioned B+ indexing scheme.



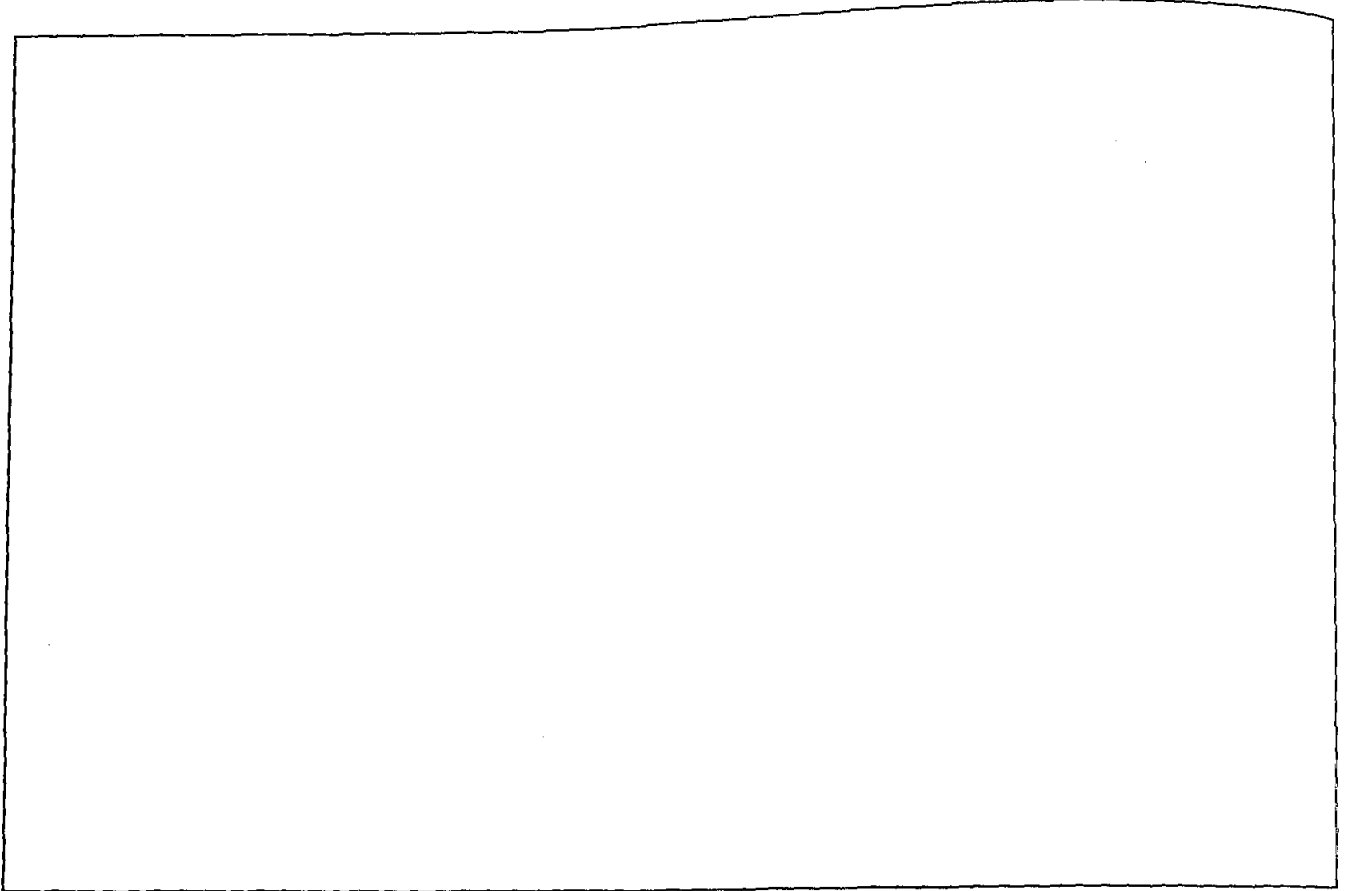
(b) Consider the following B+ tree.

[3X3]

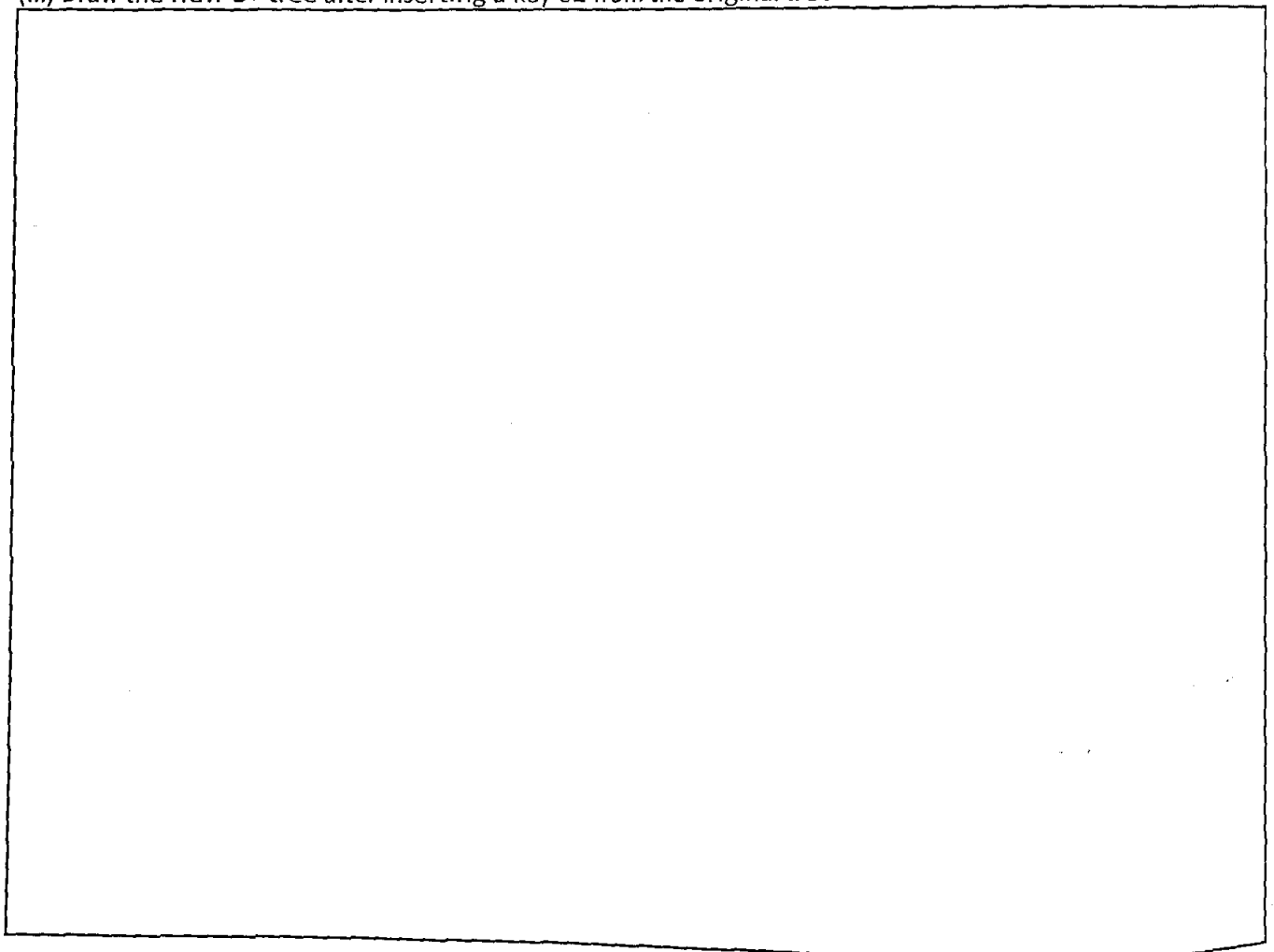


(i) Draw the new B+ tree after inserting a key 42.

(ii) Draw the new B+ tree after deleting the key 23 from the original tree.



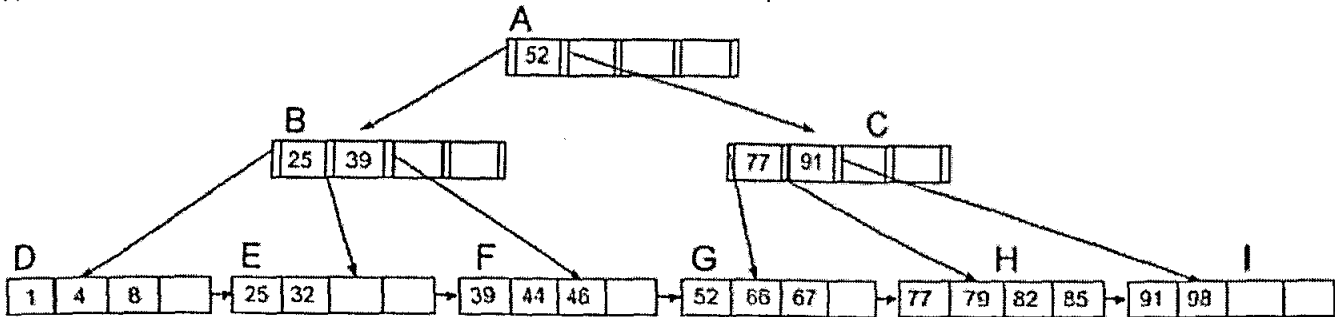
(iii) Draw the new B+ tree after inserting a key 82 from the original tree.



(c) The following table stores data of a food delivery platform: [2x4]
TwiggyFoods(orderId, driverID, customerID, origin, destination, price)

It has 1,000,000,000 records, with each page containing 100 records. We have the following assumptions: "orderId" is an integer value distributed uniformly over the range [0 - 999,999,999], "driverID" is an integer value distributed uniformly over the range [0 - 999,999], "customerID" is an integer value distributed uniformly over the range [0 - 99,999,999], "origin" and "destination" are variable-length strings with a not-null constraint and a maximum length of 100 bytes, "price" is an int value over the range [0 - 5,000].

(i) Consider the following unclustered B+ tree index on the "price" field of the relation.



Assuming that each record in the answer resides on a different page, what is the number of disk I/O required to answer the following query (explain): *SELECT * FROM TwiggyFoods WHERE price ≥ 39 AND price ≤ 78;*

(ii) For answering the following queries what is the best attribute to build a B+ tree index. Also, explain whether it should be unclustered/clustered.

SELECT T.origin, T.destination FROM TwiggyFoods T WHERE T.orderID = 45;

SELECT count() FROM TwiggyFoods T WHERE T.customerID > 5,000 and T.customerID < 7000.*

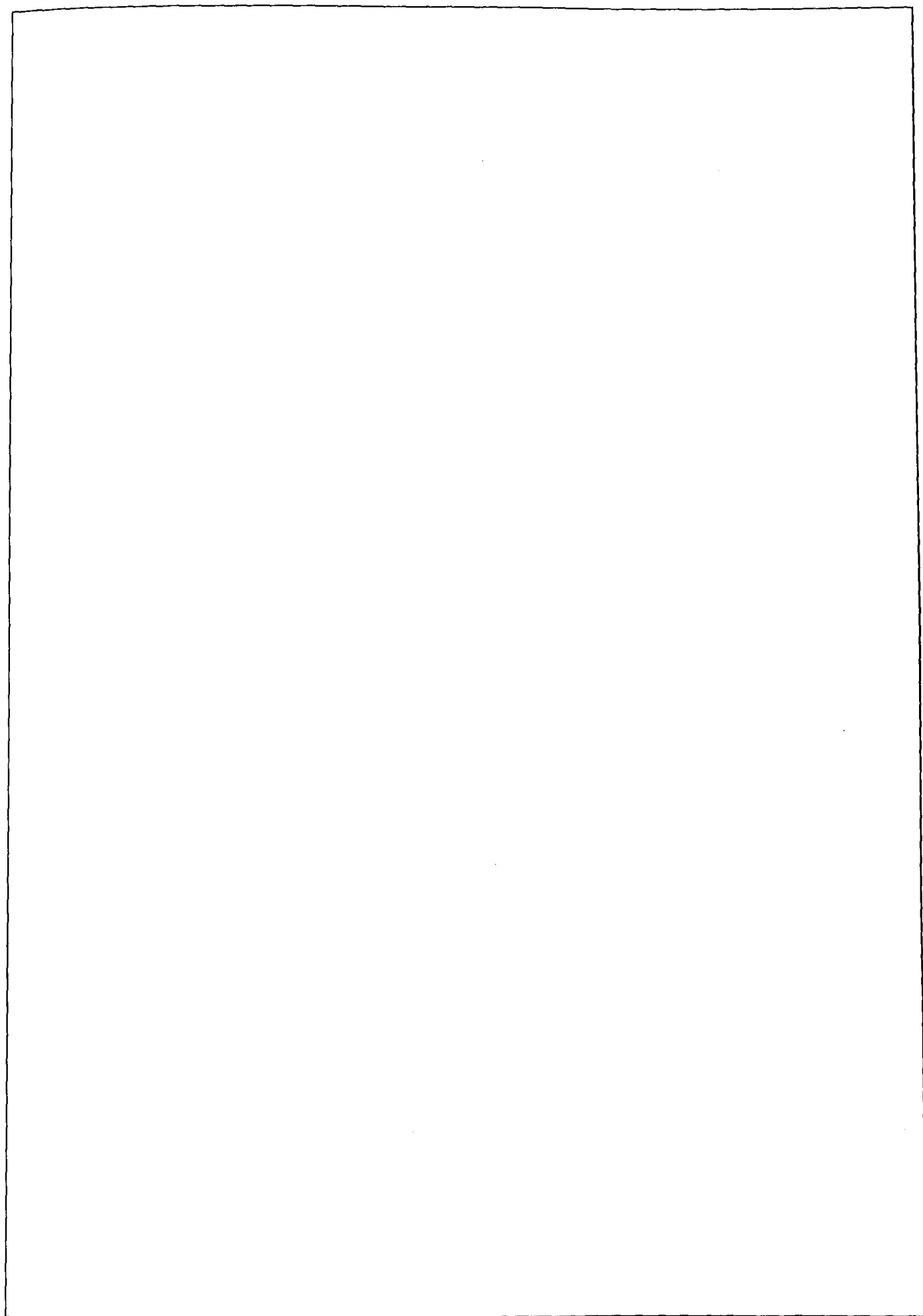
SELECT AVG(T.price) FROM TwiggyFoods T WHERE T.origin = "Sahara"

2.(a) Consider a disk with block size $B = 1024$ bytes. A block pointer is $P = 10$ bytes long. A file has $r = 1,000,000$ STUDENT records of fixed size 300 bytes. Suppose that the file is ordered by the key field `student_id` (15 bytes), and we want to construct a primary index on `student_id`. Calculate

(i) The number of first-level index entries and the number of first-level index blocks, in case of single level indexing system. Determine the number of block accesses needed to search for and retrieve a record from the file, given its `student_id` value using the single level primary index structure.

(ii) Suppose, if we want to construct an index structure on student's Aadhar number whose size is 20 bytes (one of the attributes of the record), on the above mentioned ordered file. Assume student's aadhar number is NOT NULL. Determine the number of first-level index entries and the number of first-level index blocks, in case of single level indexing system. Determine the number of block accesses needed to search for and retrieve a record from the file, given its student's Aadhar number value using the single level index structure. In case of multi-level indexing based Aadhar number, how many block accesses are required to search and retrieve a record from the file.

[3+4]



(b) Suppose you have a relation file with 10,000 pages and you have only three buffer pages. You would like to sort the file on the primary key using the external memory merge sort algorithm. [4 x 2]

(i) How many runs will you produce in the first pass? Explain.

$$\text{CEIL}(10000/3) = 3334$$

(ii) How many passes will it take to sort the file completely? Explain.

$$1 + \text{LOG}(3334) \text{ BASE } 2$$

(iii) What is the total I/O cost of sorting the file (with final write)? Explain.

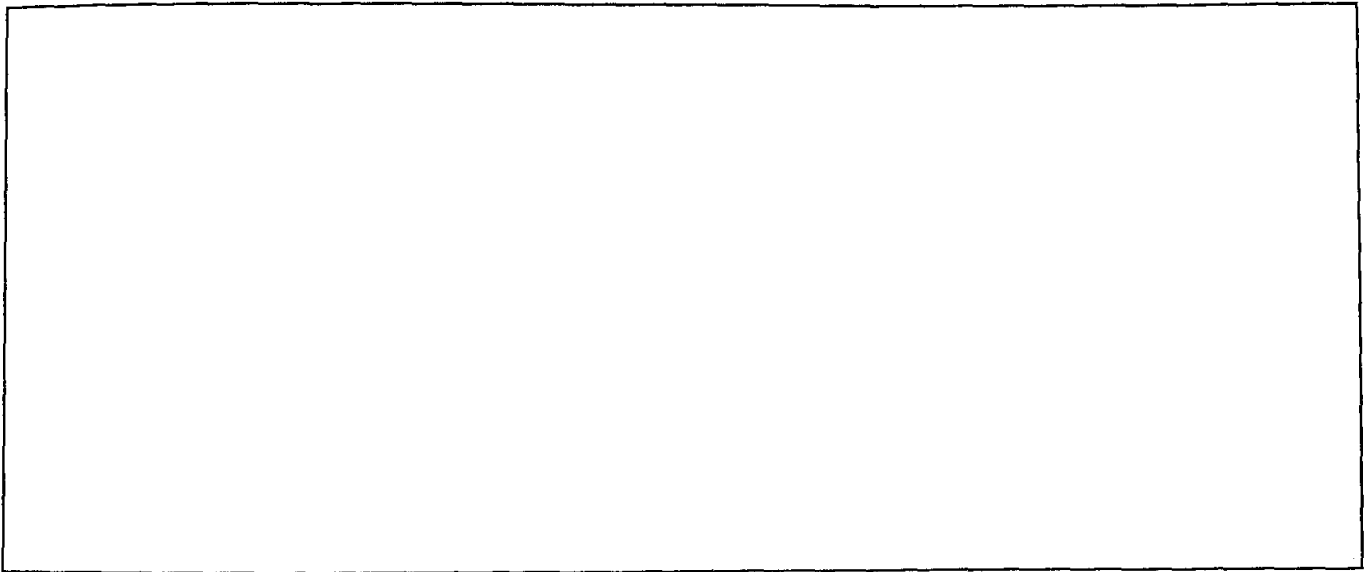
(iv) How many buffer pages do you need to sort the file completely in just two passes? Explain.

(c) The following schema and SQL query captures and reports information about a company. [2+3+5]

Emp (eid: integer, did: integer, sal: integer, hobby: char(20))
Dept (did: integer, dname: char(20), floor: integer, phone: char(10))
Finance (did: integer, budget: real, sales: real, expenses: real)

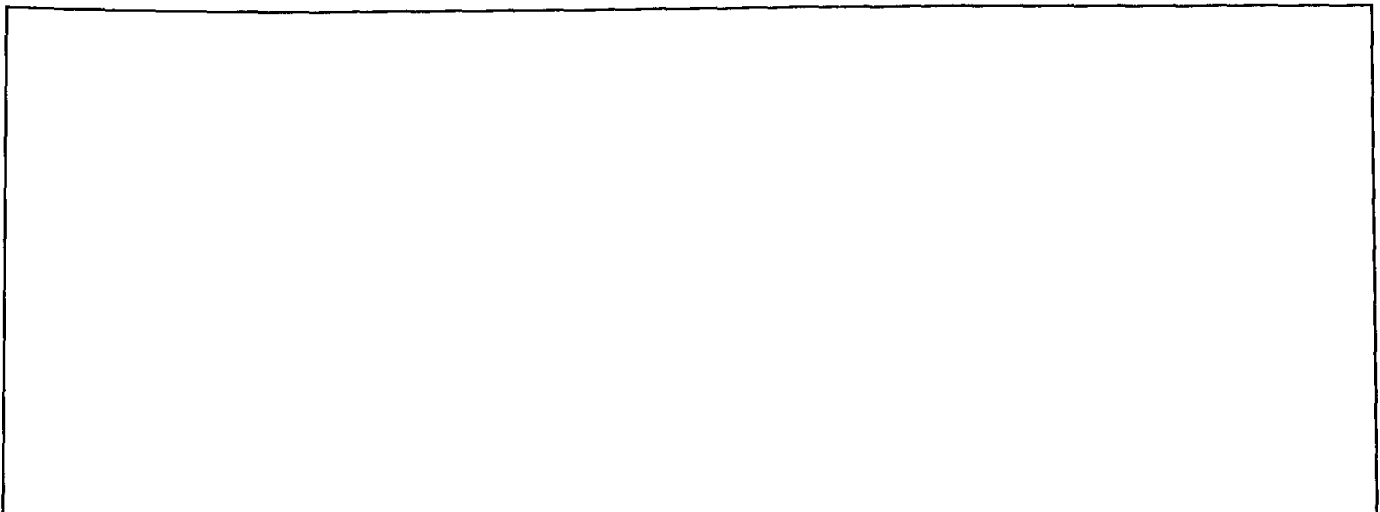
```
SELECT D.dname, F.budget
FROM Emp E, Dept D, Finance F
WHERE E.did = D.did AND D.did = F.did AND D.floor=1
AND E.sal >= 59000 AND E.hobby = 'yodelling'
```

(i) Identify a relational algebra tree that reflects the order of operations that a query optimizer would choose.



Suppose that the following additional information is available: Unclustered B+ tree indexes exist on Emp.did, Emp.sal, Dept.floor, Dept.did, and Finance.did. The system's statistics indicate that employee salaries range from 10,000 to 60,000, employees enjoy 200 different hobbies, and the company owns two floors in the building. There are a total of 50,000 employees and 5,000 departments (each with corresponding financial information) in the database.

(ii) For each of the previous SQL query's base relations (Emp, Dept and Finance) estimate the number of tuples that would be initially selected from that relation if all of the non-join predicates on that relation were applied to it before any join processing begins. Show your calculation.



(iii) Given the answer to the preceding question, and if the index nested loop join algorithm is used, estimate the cost of optimal left-deep join order in terms of total number of block reads. Show your calculations.

3.(a) When are two schedules considered to be view equivalent. Also, define view serializability, and provide an example schedule which is view serializable. When a schedule will be view serializable, but not conflict serializable? [7]

(b) Consider the following schedule for seven transactions $T_1 - T_7$. For ith transaction beginning, commit and abort are indicated by b_i , c_i and a_i

$\mathcal{T} = \{T_1, T_2, T_3, T_4, T_5, T_6, T_7\}$

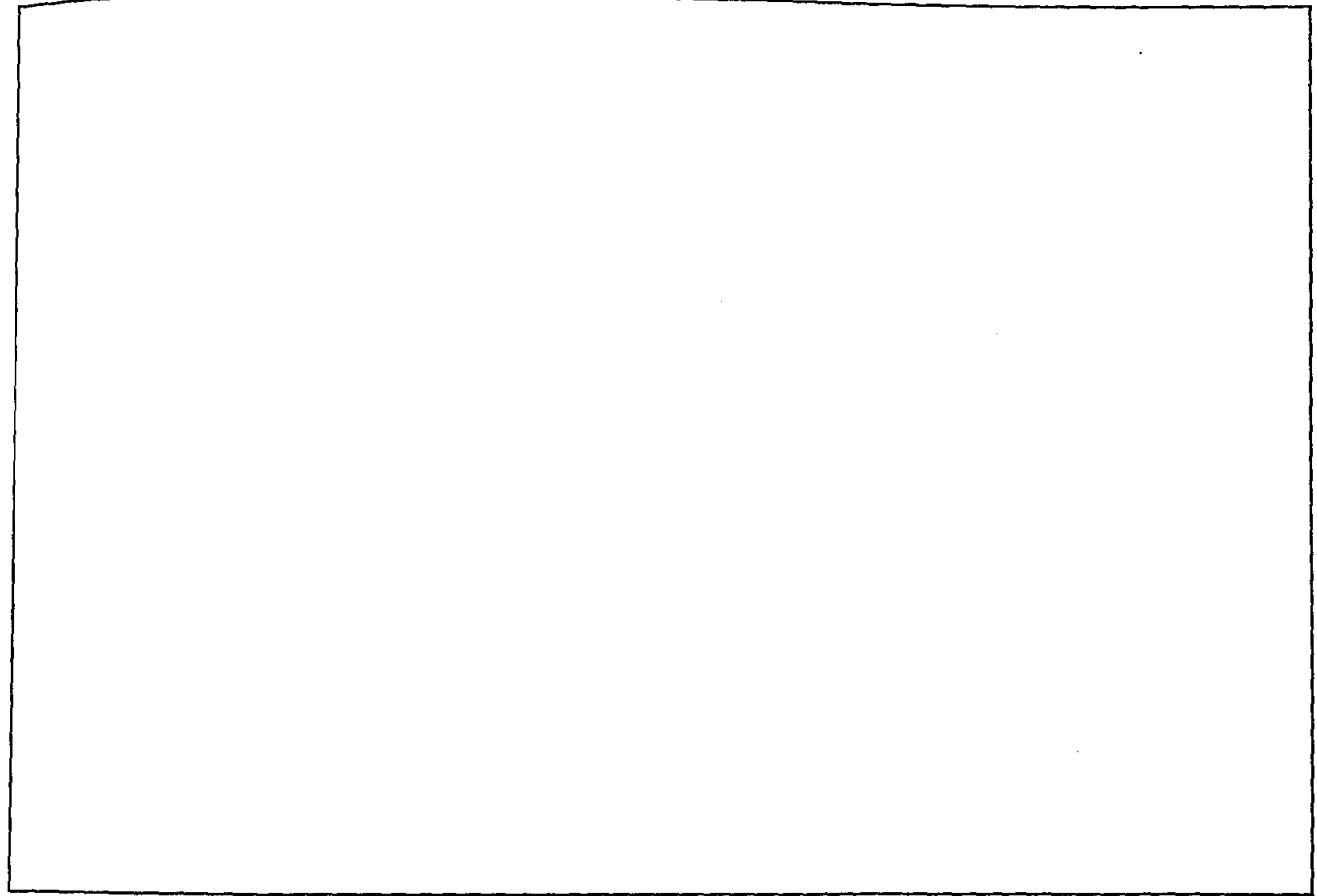
$\mathcal{H} = b_1 \rightarrow r_1(A) \rightarrow b_2 \rightarrow r_2(B) \rightarrow r_2(A) \rightarrow w_2(B) \rightarrow b_3 \rightarrow w_3(C) \rightarrow b_7 \rightarrow w_1(A) \rightarrow r_7(C) \rightarrow b_6 \rightarrow b_4 \rightarrow r_6(A) \rightarrow w_4(A) \rightarrow r_7(A) \rightarrow b_5 \rightarrow w_6(A) \rightarrow r_3(A) \rightarrow r_7(B) \rightarrow r_6(B) \rightarrow r_7(C) \rightarrow w_5(D) \rightarrow w_6(C) \rightarrow a_6 \rightarrow a_1 \rightarrow r_5(B) \rightarrow r_2(C) \rightarrow r_5(C) \rightarrow c_2 \rightarrow r_3(C) \rightarrow w_5(C) \rightarrow c_3 \rightarrow w_5(B) \rightarrow c_7 \rightarrow c_5 \rightarrow w_4(D) \rightarrow c_4.$

(i) Draw the precedence graph of the committed transactions of the above schedule.

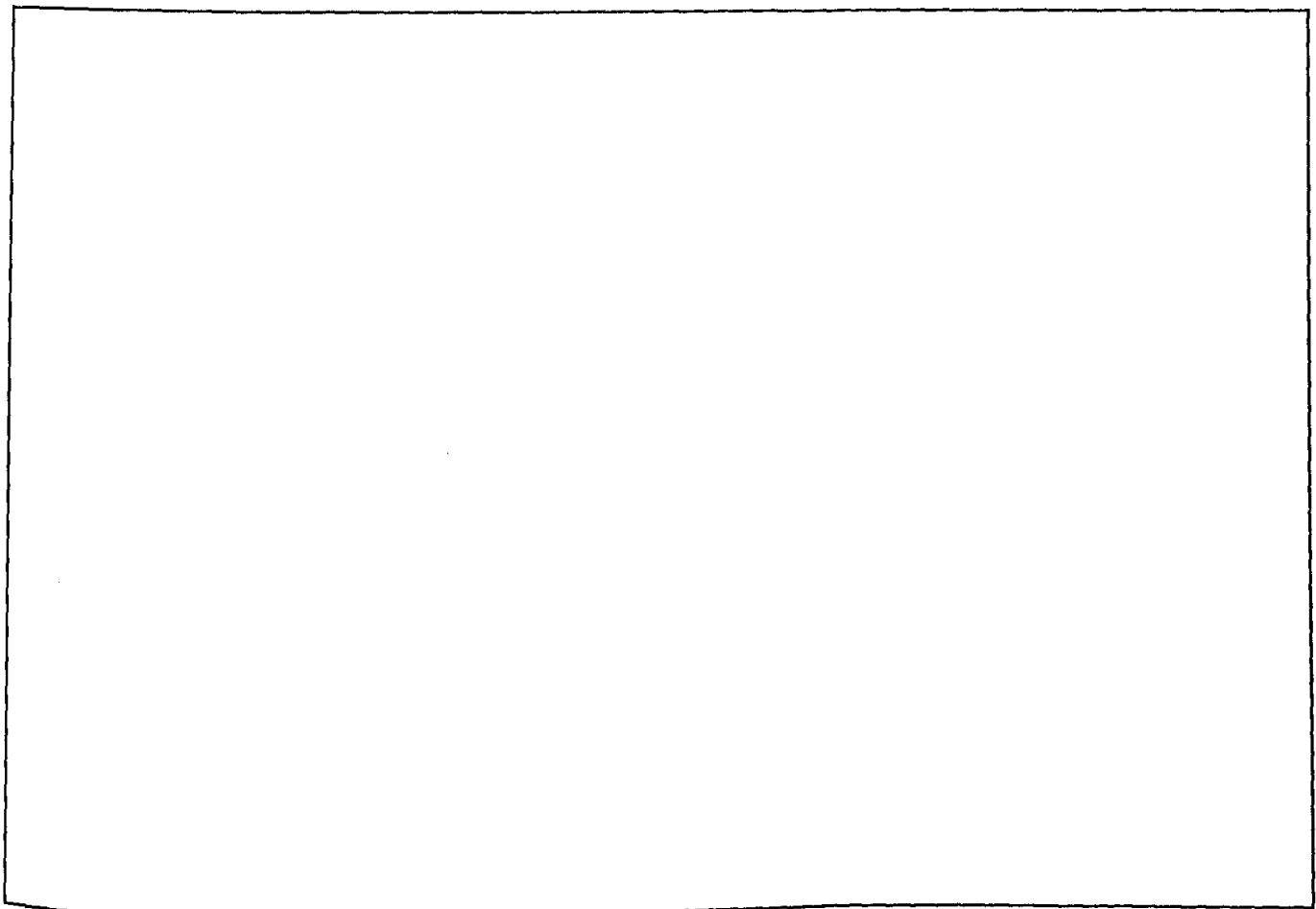
[4]

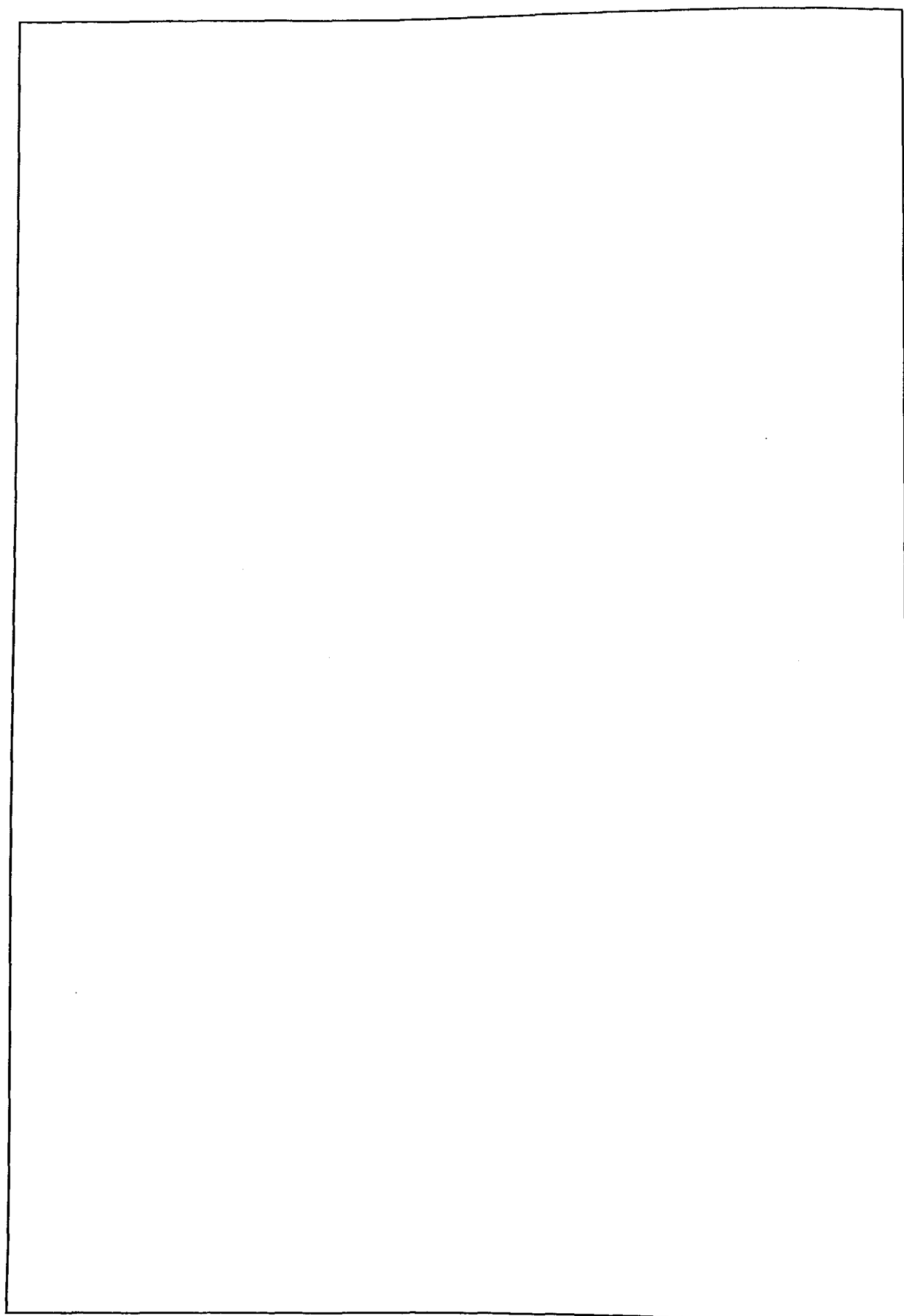
(ii) State which of the following properties holds for the schedule: conflict serializable, recoverable, avoids cascading rollback. Explain. [4]

(c) Suppose a schedule consisting of 5 transactions is as follows: w1(a) r2(a) w1(b) w3(c) r2(c) r4(b) w2(d) w4(e) r5(d) w5(e). Here the notation w1(a): write operation on data item "a" by transaction 1. Similarly, "r" refers to read operation. Comment on conflict serializability of this schedule. Mention all possible serial schedules, in case if it is conflict-serializable. [5]



(d) Briefly, explain the isolation levels specified by SQL standard. What is snap-shot isolation ? [5]





4(a) Consider the following two transactions.

[3.5+1.5]

$T1 = r1(A) w1(A) r1(B) w1(B);$ $T2 = r2(B) r2(A) w2(A) w2(B)$

(i) Add exclusive-lock (lock-X), unlock (U), and COMMIT, instructions in the transactions such that any of the allowed concurrent schedule is - conflict serializable, may lead to deadlock, but avoids cascading rollback. Explain your answer.

(ii) If a deadlock occurs for the schedule in previous question the *wait-die* deadlock prevention scheme is used. If T1 starts first, which transaction is rolled-back under this scheme? Explain.

(b) Explain the salient features of tree protocol in the context of concurrency control. Mention the advantages and disadvantages of tree protocol compared to two-phase locking protocol. [5]

(c) In timestamp ordering protocol, state the conditions that needs to be checked before read and write operations.

[5]

(d) Consider a timestamp based concurrency controller. Below are several sequences of events, including start events, where sti means that transaction T_i starts and coi means T_i commits. These sequences represent real time, and the timestamp-based scheduler will allocate timestamps to transactions in the order of their starts. In each case below, say what happens with the last request. You have to choose between one of the following three possible outcomes: the request is 1. accepted, 2. ignored, 3. transaction is rolled back. Explain the reason for the mentioned outcome. [5]

(i) $st1; st2; r1(A); r2(A); w1(B); w2(B)$

(ii) $st1; st2; r2(A); co2; r1(A); w1(A)$

(iii) $st1; st2; st3; r1(A); w3(A); co3; r2(B); w2(A)$

(iv) $st1; st2; st3; r1(A); w1(A); w3(A); r2(A)$

(v) $st1; st2; st3; r1(A); w2(A); w3(A); r2(A)$

(e). Consider the log records written by transactions T1 and T2: <START T1>; <T1, A, 10, 11>; <START T2>; <T2, B, 20, 21>; < T1,C, 30, 31>; <T2, D, 40, 41>; <COMMIT, T2>; <T1, E, 50, 51>; CRASH <COMMIT, T1>

What are the changes made by the recovery manager to the disk and the log during recovery from crash, if the last log record to appear on disk is <T1, E, 50, 51>.

[5]

ANS: Database Changes:	
A:	
B:	
C:	
D:	
E:	
ANS: Log Record Changes:	

ROUGH WORK